

Optimization Techniques in Power System (BT EE610)



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Unit-III

Ant Colony Optimization

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What is Ant Colony Optimization?

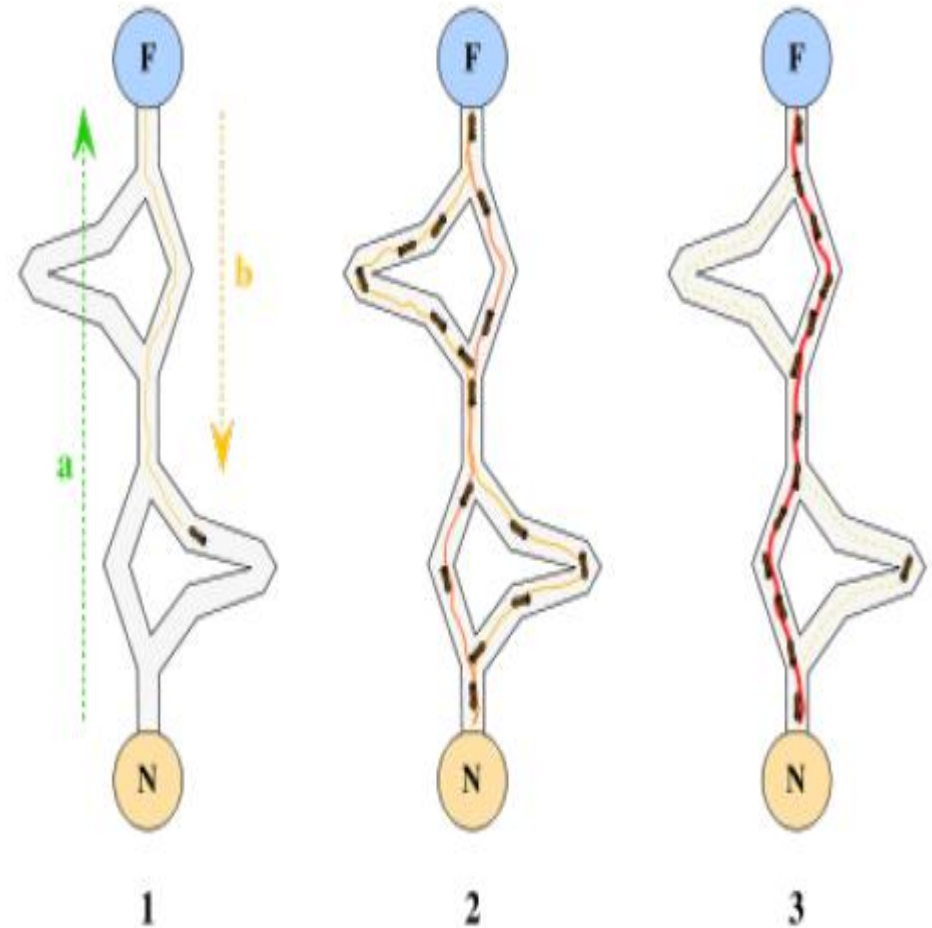
ACO

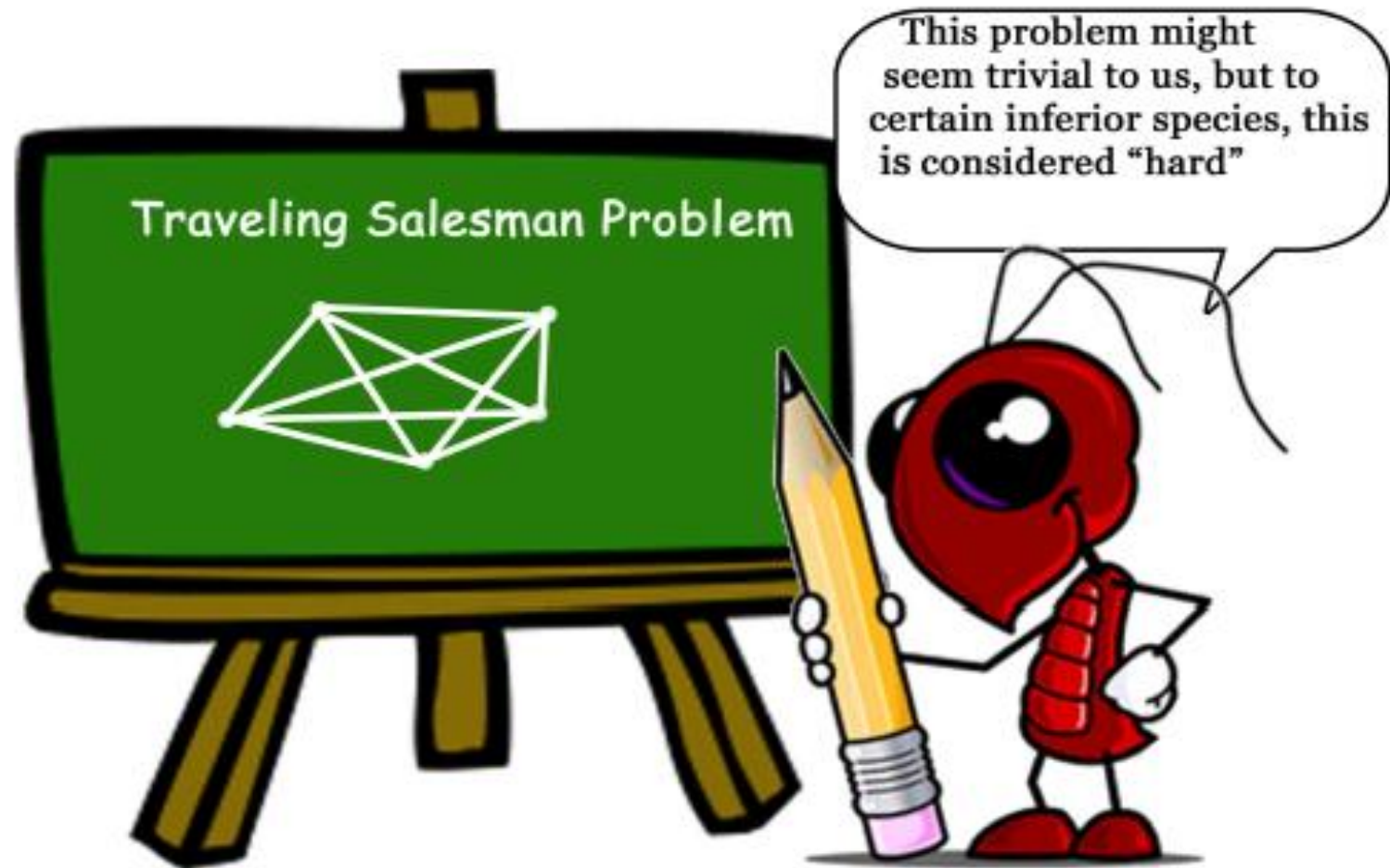
- Probabilistic technique.
- Searching for optimal path in the graph based on behaviour of ants seeking a path between their colony and source of food.
- Meta-heuristic optimization

ACO Concept

Overview of the Concept

- Ants navigate from nest to food source. **Ants are blind!**
- Shortest path is discovered via **pheromone** trails.
- Each ant moves at random
- Pheromone is deposited on path
- More pheromone on path increases probability of path being followed





Art by Pratik

ACO System

Overview of the System

- Virtual trail accumulated on path segments
- Path selected at random based on amount of "trail" present on possible paths from starting node
- Ant reaches next node, selects next path
- Continues until reaches starting node
- Finished tour is a solution.
- Tour is analyzed for optimality

Meta-heuristic

- Heuristic method for solving a very general class of computational problems by combining user-given heuristics in the hope of obtaining a more efficient procedure.
- **ACO is meta-heuristic**
- **Soft computing technique for solving hard discrete** optimization problems

History

- Ant System was developed by Marco Dorigo (Italy) in his PhD thesis in 1992.
- **Max-Min Ant System developed by Hoos and Stützle in 1996**
- **Ant Colony was developed by Gambardella Dorigo in 1997**

The ACO Meta-heuristic

ACO

Set Parameters, Initialize pheromone trails

SCHEDULE ACTIVITIES

- 1 Construct Ant Solutions
- 2 Daemon Actions (optional)
- 3 Update Pheromones

Virtual trail accumulated on path segments

ACO - Construct Ant Solutions



ACO - Construct Ant Solutions

An ant will move from node i to node j with probability

$$p_{i,j} = \frac{(\tau_{i,j}^{\alpha})(\eta_{i,j}^{\beta})}{\sum (\tau_{i,j}^{\alpha})(\eta_{i,j}^{\beta})}$$

where

$\tau_{i,j}$ is the amount of pheromone on edge i,j

α is a parameter to control the influence of $\tau_{i,j}$

$\eta_{i,j}$ is the desirability of edge i,j (typically $1/d_{i,j}$)

β is a parameter to control the influence of $\eta_{i,j}$

ACO - Pheromone Update

ACO - Pheromone Update

Amount of pheromone is updated according to the equation

$$\tau_{ij} = (1 - \rho)\tau_{ij} + \Delta\tau_{ij}$$

where

τ_{ij} is the amount of pheromone on a given edge i, j

ρ is the rate of pheromone evaporation

$\Delta\tau_{ij}$ is the amount of pheromone deposited, typically given by

$$\Delta\tau_{ij}^k = \begin{cases} 1/L_k & \text{if ant } k \text{ travels on edge } i, j \\ 0 & \text{otherwise} \end{cases}$$

where L_k is the cost of the k^{th} ant's tour (typically length).

ACO - Ant System

ACO - Ant System

- First ACO algorithm to be proposed (1992)
- Pheromone values are updated by all the ants that have completed the tour.

$$\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij} + \sum_{k=1}^m \Delta\tau_{ij}^k,$$

where

ρ is the evaporation rate

m is the number of ants

$\Delta\tau_{ij}^k$ is pheromone quantity laid on edge (i, j) by the k^{th} ant

$$\Delta\tau_{ij}^k = \begin{cases} 1/L_k & \text{if ant } k \text{ travels on edge } i, j \\ 0 & \text{otherwise} \end{cases}$$

where L_k is the tour length of the k^{th} ant.

ACO - Ant Colony System

ACO - Ant Colony System

- First major improvement over Ant System
- Differences with Ant System:
 - 1 Decision Rule - Pseudorandom proportional rule
 - 2 Local Pheromone Update
 - 3 Best only offline Pheromone Update
- Ants in ACS use the **pseudorandom proportional rule**
- Probability for an ant to move from city i to city j depends on a random variable q uniformly distributed over $[0, 1]$, and a parameter q_0 .
- If $q \leq q_0$, then, among the feasible components, the component that maximizes the product $\tau_{ij}\eta_{ij}^\beta$ is chosen, otherwise the same equation as in Ant System is used.
- This rule favours exploitation of pheromone information

ACO - Ant Colony System

- Diversifying component against exploitation: **local pheromone update**.
- The local pheromone update is performed by all ants after each step.
- Each ant applies it only to the last edge traversed:

$$\tau_{ij} = (1 - \varphi) \cdot \tau_{ij} + \varphi \cdot \tau_0$$

where

$\varphi \in (0, 1]$ is the pheromone decay coefficient

τ_0 is the initial value of the pheromone (value kept small

Why?)

ACO - Ant Colony System

- Best only offline pheromone update after construction
- Offline pheromone update equation

$$\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij} + \rho \cdot \Delta\tau_{ij}^{best}$$

where

$$\tau_{ij}^{best} = \begin{cases} 1/L_{best} & \text{if best ant } k \text{ travels on edge } i,j \\ 0 & \text{otherwise} \end{cases}$$

- L_{best} can be set to the length of the best tour found in the current iteration or the best solution found since the start of the algorithm.

Application of ACO

ACO

- Routing in telecommunication networks
- Traveling Salesman
- Graph Coloring
- Scheduling
- Constraint Satisfaction

ACO - MAX-MIN Ant System

- Differences with Ant System:
 - 1 Best only offline Pheromone Update
 - 2 Min and Max values of the pheromone are explicitly limited
 - τ_{ij} is constrained between τ_{min} and τ_{max} (explicitly set by algorithm designer).
 - After pheromone update, τ_{ij} is set to τ_{max} if $\tau_{ij} > \tau_{max}$ and to τ_{min} if $\tau_{ij} < \tau_{min}$

Merit of ACO

ACO

- Inherent parallelism
- Positive Feedback accounts for rapid discovery of good solutions
- Efficient for Traveling Salesman Problem and similar problems
- Can be used in dynamic applications (adapts to changes such as new distances, etc)

Demerit of ACO

ACO

- Theoretical analysis is difficult
- Sequences of random decisions (not independent)
- Probability distribution changes by iteration
- Research is experimental rather than theoretical
- Time to convergence uncertain (but convergence is guaranteed!)

References

- M. Dorigo, M. Birattari, T. Stützle, “Ant Colony Optimization – Artificial Ants as a Computational Intelligence Technique”, IEEE Computational Intelligence Magazine, 2006
- M. Dorigo K. Socha, “An Introduction to Ant Colony Optimization”, T. F. Gonzalez, Approximation Algorithms and Metaheuristics, CRC Press, 2007
- M. Dorigo T. Stützle, “The Ant Colony Optimization Metaheuristic: Algorithms, Applications, and Advances”, Handbook of Metaheuristics, 2002